Implementation of ETL for Video Game Sales Data Analysis Using pentaho Data Integration

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Abstract—

Over the past few years, the gaming industry has continued to expand and generate a large volume of sales records from different publishers, genres, and platforms. This growing amount of data creates opportunities for market analysis, trend identification, and performance evaluation. However, the raw form of this data is often not ready for direct analysis because it may contain duplicate entries, incomplete information, and inconsistent formatting.In this project, an ETL-based data pipeline was developed to process video game sales data into a more structured format. The dataset was collected from Kaggle and processed using Pentaho Data Integration. Throughout the pipeline, the data went through extraction, cleaning, validation, and transformation before being stored in a MySQL database. A star schema model was then applied to improve analytical efficiency. The final data output was visualized using Power BI to present sales insights in a clearer and more interactive way.Based on the implementation results, the ETL process successfully improved data consistency and reduced redundancy, making the dataset easier to analyze and interpret. This project demonstrates how ETL and data warehousing can support better business intelligence practices and data-driven decisions [1], [3], [9].

keyword:ETL, Data Warehouse, Pentaho Data Integration, Star Schema, Video Game Sales, Power BI, Business Intelligence

I.INTRODUCTION

The gaming market has experienced strong growth in recent years and has become one of the most active sectors in digital entertainment. With thousands of games being released globally, sales records continue to increase and produce a large amount of data. This data can be useful for understanding consumer preferences, measuring product success, and analyzing market patterns [2].Even though the data provides valuable insights, raw datasets are not always easy to use directly. During this project, several issues were found in the original dataset, such as repeated records, inconsistent naming formats, and missing publisher information. These issues could affect analysis accuracy if left untreated.Previous studies explain that ETL pipelines play an important role in transforming raw datasets into cleaner and more organized structures [1], [3]. In addition, the use of star schema in data warehousing has been proven to improve analytical performance and simplify reporting processes [4], [5].Therefore, this project was designed to process video game sales data using Pentaho Data Integration and transform it into an analytical dataset that can later be visualized through Power BI dashboards.

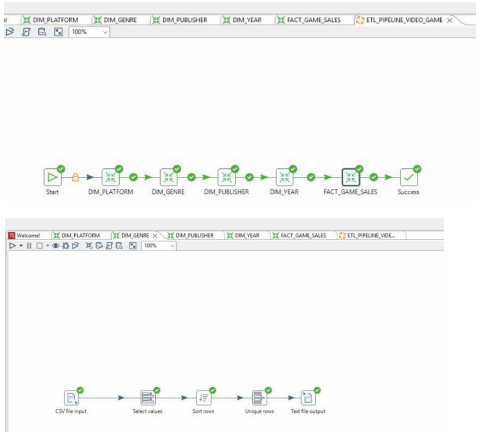
II.Data Source

The dataset used in this project was obtained from Kaggle and contains historical sales data of video games from multiple platforms. The data includes attributes such as game title, release year, platform, genre, publisher, and total global sales. This dataset was selected because it contains enough variation for performing data analysis and building a complete ETL workflow. It also allows the implementation of multidimensional modeling by separating the data into several analytical categories.The dataset consists of thousands of records, making it suitable for demonstrating data engineering concepts such as ingestion, transformation, and warehouse modeling [2].

III.Data Ingestion

At the ingestion stage, the dataset was loaded into Pentaho Data Integration using a full-load approach. This method imports the entire CSV file into the ETL system in one execution.A full-load strategy was chosen because the dataset is static and does not receive regular updates. This makes the process simpler and more efficient for this project.The extracted data then became the main source for all transformation activities. This stage ensures that all original records are captured before any modifications are applied [3], [8]

Figure 1. Pentaho ETL Workflow



IV. Data Strorage

After the ingestion process was completed, the data storage phase was divided into two separate layers: raw storage and processed storage. This approach was used to maintain data integrity and improve data management throughout the ETL pipeline.The raw storage layer contains the original CSV file exactly as it was obtained from Kaggle. Keeping the original file unchanged is important because it acts as the backup source in case the transformation process needs to be repeated. This also allows the team to verify the original data whenever inconsistencies are found.The processed storage layer is the final destination of the transformed dataset. In this project, MySQL was used as the main storage system because it provides structured table management and efficient query execution. Once the ETL process was completed, the cleaned data was loaded into MySQL and separated into fact and dimension tables.This two-layer storage model improves workflow flexibility because it separates unprocessed data from analytical data. Similar approaches have been widely used in modern data warehouse implementations to improve reliability and data accessibility [1], [7].

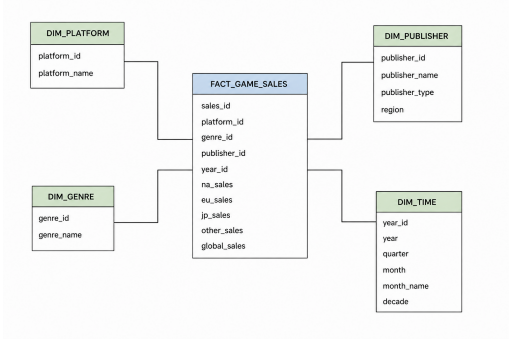
V. Data Transformation

The transformation stage is one of the most important parts of the ETL pipeline because it converts raw data into a structured and usable format. In this project, multiple transformation activities were carried out to improve the dataset before storing it in the data warehouse.The first transformation step focused on data cleaning. This included removing duplicate rows, identifying missing values, and correcting inconsistent text formatting. For example, some publisher names appeared with different capitalization styles, which could lead to duplicate dimension entries if not standardized.Another transformation process involved converting data types into the correct format. The release year column was converted into numeric values, while sales values were checked to ensure they were stored as decimal numbers.After the cleaning stage was completed, the data was normalized into smaller dimension tables. The dataset was divided into platform, genre, publisher, and year dimensions. This separation reduces redundancy and improves data consistency.Finally, the fact table was created to store the main sales metrics and connect all dimension tables using foreign key relationships. This structure improves query performance and supports analytical operations such as aggregation and filtering [3], [5], [8].The transformation phase also highlighted the importance of handling small inconsistencies carefully, since even minor errors in raw data could affect dashboard accuracy later.

VI. Data Modeling

To organize the transformed dataset for analysis, this project used the star schema model. The star schema is one of the most commonly used data warehouse designs because of its simplicity and efficiency.The model consists of one central fact table containing sales values and several dimension tables containing descriptive information. The fact table stores measurable values such as total sales, while the dimension tables provide context such as game title, platform, genre, publisher, and release year.This structure makes analytical queries faster because dimension tables can be joined directly to the fact table without complex relationships. Compared to normalized transactional databases, star schema reduces query complexity and improves reporting performance [4], [5].In this project, the star schema also made it easier to integrate the dataset into Power BI since the relationship structure was already clear and organized.

Figure 2. Star Schema Design



VII. Data Quality and Validation

Data validation was performed to ensure that the transformed dataset was accurate and reliable before loading it into MySQL. Data quality is essential because poor-quality data can lead to incorrect analysis and misleading business insights.Several validation rules were applied during this phase. First, sales values were checked to ensure they were greater than zero. Records with zero or invalid sales values were reviewed to avoid incorrect calculations.Second, the release year column was validated to confirm that all values were numeric and within a reasonable range. This was important for yearly trend analysis.Third, publisher and genre columns were checked for missing or empty values. These fields were important because they represent major analytical dimensions.Through these validation steps, the dataset became cleaner, more accurate, and more reliable for dashboard creation [1], [5].

VIII. Workflow Automation and Orchestration

To make the ETL process more efficient, Pentaho Job was used to automate the workflow. Instead of running each transformation step manually, all stages were connected into one automated pipeline.The workflow begins with extracting data from the CSV file, followed by data cleaning, transformation, validation, and loading into MySQL. Each step was linked sequentially so that the process could run automatically in the correct order.Automation reduces human error and saves time, especially when the workflow needs to be executed multiple

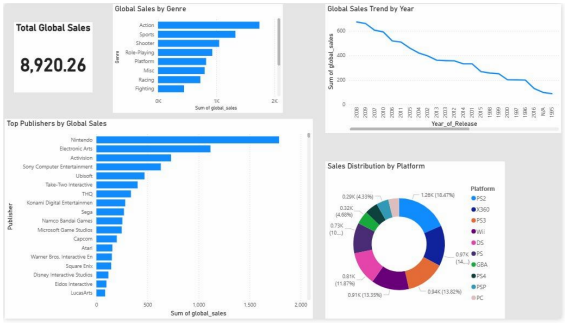
times. It also makes the pipeline easier to maintain because changes can be managed within one centralized job structure [3], [9].In real-world applications, automation is important for improving consistency and ensuring that the same logic is applied in every execution.

IX. Data Consumption and Visualization

To make the ETL process more efficient, Pentaho Job was used to automate the workflow. Instead of running each transformation step manually, all stages were connected into one

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Figure 3. Power BI Dashboard Visualization



X. Evaluation and Reflection

This project showed that building an ETL pipeline using Pentaho Data Integration can significantly improve how raw datasets are processed and organized. The visual workflow provided by Pentaho made it easier to design and monitor each stage of the process.One important observation during the project was that data cleaning required more attention than expected. Small inconsistencies such as duplicated records or inconsistent text formatting had a noticeable impact on the final analysis.The star schema model proved to be effective because it improved data organization and simplified analytical queries. This made the Power BI integration smoother and more efficient.However, there are still limitations in the current implementation. Since the dataset is static, it cannot reflect live market changes. The pipeline also only supports CSV files and does not yet connect to external APIs.Future improvements could include integrating real-time data sources, cloud-based storage systems, and automated dashboard refresh mechanisms to improve scalability and usability [7], [10].

XI. Conclusion

This project successfully developed an ETL-based pipeline for processing video game sales data from raw format into a structured analytical database. Through Pentaho Data Integration, the dataset was extracted, cleaned, transformed, validated, and loaded into MySQL.The use of a star schema model improved data organization and made analytical queries more efficient. Power BI dashboards then transformed the processed data into meaningful visual insights, making it easier to analyze sales performance across genres, platforms, and time periods.Overall, this project demonstrates that ETL pipelines and data warehouse architectures are highly useful for improving data quality and supporting business intelligence processes. The implementation also provides practical experience in applying data engineering concepts to real-world datasets [1], [5], [9].

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Fig. 1.